
Algorithmen und Wahrscheinlichkeit

Exercise Session 3

Exercise 1 – *Finding All Augmenting Path*

Let $G = (A \cup B, E)$ be a connected bipartite graph, and let M be a matching, i.e., a set of edges such that every vertex is the endpoint of at most one edge in M . Assume that M is not cardinality-maximal. In the lecture, you saw how to find the shortest augmenting path to M in G (see script or slides for a formal definition of augmenting paths) in time $O(|V| + |E|)$.

For the Hopcroft-Karp algorithm, it was claimed that we can find an inclusion-maximal collection of disjoint shortest augmenting paths in time $O(|V| + |E|)$. Describe an algorithm doing this.